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R Language for Biological Data Analysis

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What do you know about R?



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Go to menti.com and
type code 7965 3394!

Lets see what you
already know about R!

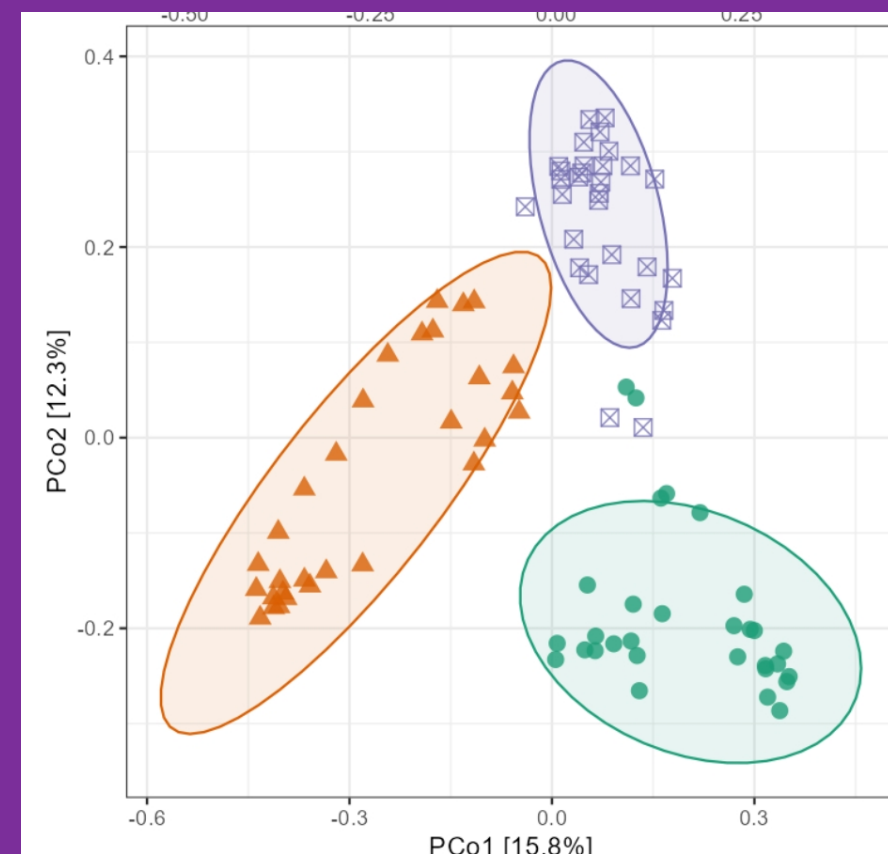
Why R is suitable for bioinformatician?



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In the simplest language, R is actually a calculator that is able to remember things and draw!

But, why R is popular in bioinformatics?



Why I need to use R compared to Excel instead?

SCIENCE REPORT TECH

Scientists rename human genes to stop Microsoft Excel from misreading them as dates



Illustration by Alex Castro / The Verge

/ Sometimes it's easier to rewrite genetics than update Excel

by [James Vincent](#)
Aug 6, 2020 at 7:44 PM GMT+7

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There is a gene which is currently called as SEPTIN1, which previously is called as SEPT1.

Excel misreads it as 1 September instead!

I think Python also can be used. Why I need to use R?



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Actually, R and Python in bioinformatics are complementary to each other. R and Python usage in bioinformatics can be summarized as follows:

	R	Python
Strongest at	statistics, visualization, genomics-specific analysis	general-purpose scripting, pipelines, machine learning
Field-specific ecosystem	Bioconductor	Biopython (smaller, more general)
Typical use in this field	differential expression, statistical modelling, publication-ready figures	workflow automation (Snakemake/Nextflow), large-scale data engineering
Learning curve for a non-programmer	gentler for tabular/statistical thinking	gentler if you'll also write general software

Can I use R as executable script or even standalone app?

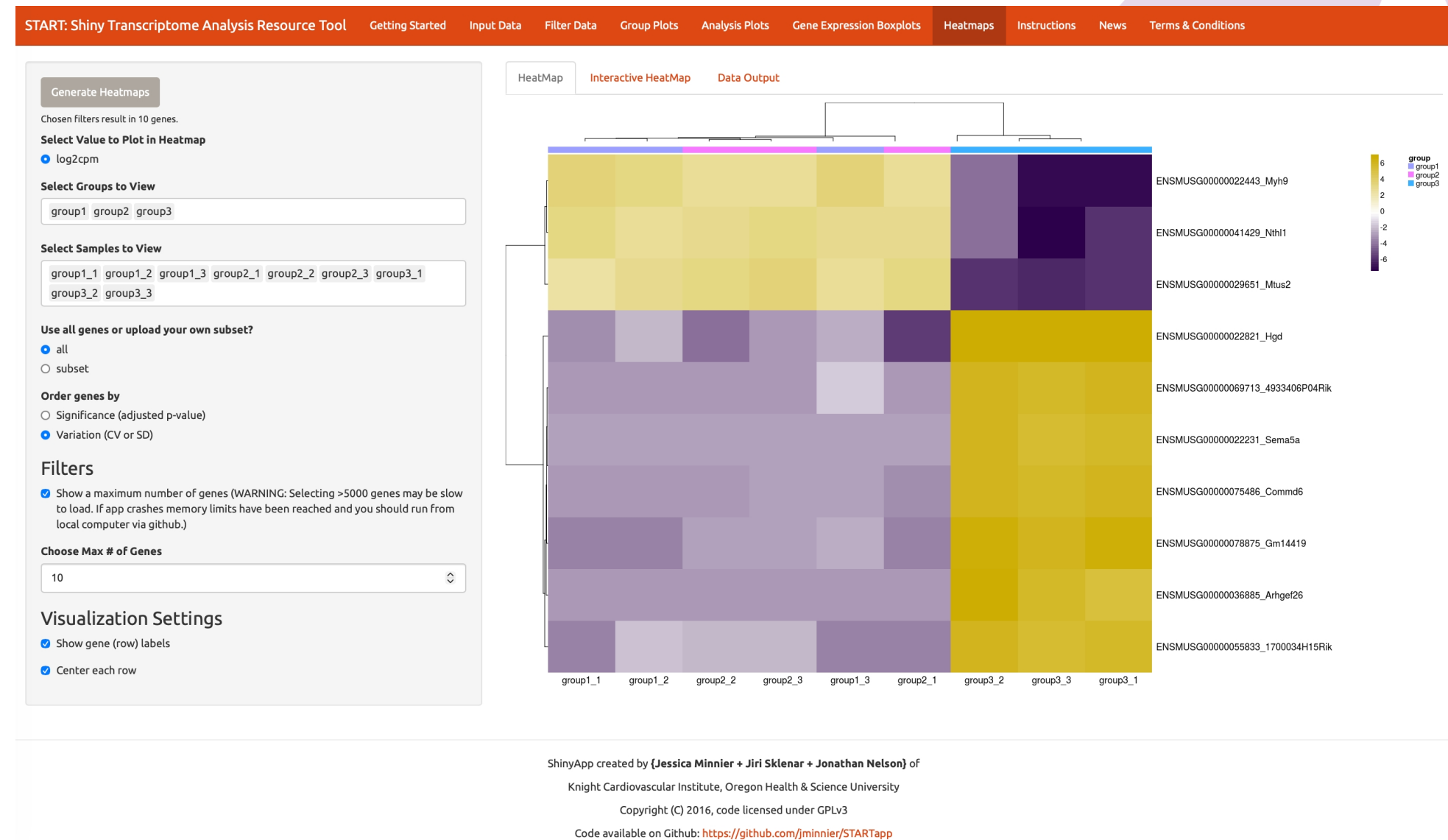


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Yes you can!

As a bioinformatician, we usually use Bash as orchestrator and R/Python as the executor script.

This is very important when you work in cloud computing and HPC.



<https://kcv.shinyapps.io/START/>

Any example of R-based tools for bioinformatics?



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Genomics



vegan: an R package for community ecologists

Ordination methods, diversity analysis and other functions for community and vegetation ecologists.

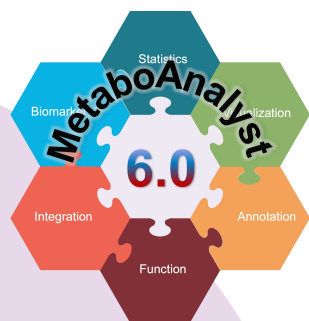
Transcriptomics



Proteomics



Metabolomics and Constraint-Based Modeling



sybil - Efficient Constrained Based Modelling in R

The package `sybil` is a Systems Biology Library for R, implementing algorithms for constraint based analyses of metabolic networks (e.g. flux-balance analysis (FBA), minimization of metabolic adjustment (MOMA), regulatory on/off minimization (ROOM), robustness analysis and flux variability analysis).

Since R is a versatile language, you can apply R for any kind of bioinformatics work.

Even it is in the upstream, such as FASTQ data processing, or in downstream such as visualization.

Many tools then! How I choose it? Can you give me the tips and trick?



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1. Tools are evolving, but fundamental knowledge stays. Prioritize biological fundamentals before choosing any tools!
2. There is no kind of silver bullet in bioinformatics. All tools have pros and cons, which usually already written in their docs/vignette. READ IT CAREFULLY!
3. Start from the most familiar analysis for any works, rather than choose the advanced one. Make people understand what we already analyzed!

How about learning R as programming language? Is there any tips?



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1. R was built from tabular data, which is quite familiar for majority of people, including people with life-science background.
2. No need to memorize all command. You will remember it as your hand types those.
3. No need to worry when you got the error.
4. Learn Bash and Python are a combo that made us, bioinformatician, formidable!

How about the structure of hands-on session today?



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R Introduction

This session introduce you to R and RStudio, data type in R, how to import and manipulation data in R, and get pretty visualization using ggplot based on synthetic data of cosmetic testing measurement.

R for Omics Dataset

This session introduce you to use Phyloseq, a dedicated package for microbiome data, to obtain taxabar plot, alpha diversity and beta diversity based on data from Hulpusch et al (2023).

Automation using R

This session introduce you to make automation script using function in R, for making reproducible script!



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R you ready to rock?

Let's begin!